9. Analytical Methodology

Design

This project uses descriptive and exploratory methods. The analysis summarizes the supplied dataset and does not establish causal effects.

Analysis stages

1. **Structural profiling:** row count, fields, types, distinct values, nulls, uniqueness.
2. **Univariate analysis:** distributions of price, range, battery capacity, efficiency, charging, ratings, and sales.
3. **Group comparisons:** category, region, manufacturer, battery type, use case, and model year where appropriate.
4. **Relationship analysis:** correlation and scatterplots for numeric fields.
5. **Visualization:** KPI cards, sorted bars, histograms/box plots, scatterplots, and year/category comparisons.
6. **Validation:** reconcile aggregations and check the effects of nulls and extreme values.
7. **Communication:** present findings with definitions, filters, and caveats.

Statistical choices

- Mean is useful but sensitive to outliers; pair it with median for skewed metrics.
- Use counts to show dataset composition, not market share unless sampling and record meaning justify that interpretation.
- Pearson correlation describes linear association and can be distorted by outliers or generated relationships.
- Use box plots or percentiles to communicate spread.
- Compare year groups cautiously because the file's row count per year is fairly balanced by construction or sampling; this alone does not imply annual market volume.

Reproducibility

Every chart should record source fields, aggregation, filters, null exclusions, and any calculated field. The same filters and metric definitions must be used in the report and demo.